

Credit Card Portfolio Risk Review

Decision requested: define follow-up objectives and review capacity, then validate both candidate scopes independently.



EXECUTIVE FINDINGS

Risk concentration: PAY_0 >= 2 selects 10.43% of clients and captures 32.81% of positive labels, at a 69.55% label rate. However, 67.19% of positives remain outside this group.

Limits behave differently by status: rates decline with higher limits within PAY_0 <= 0 and = 1, but remain around 69%-71% within >= 2.

Broader scope: expanding to PAY_0 > 0 raises review volume from 3,130 to 6,818 and coverage from 32.81% to 51.67%.

CANDIDATE REVIEW SCOPES

Scope	Clients	Workload share	Positive labels	Coverage	Label rate
A: PAY_0 >= 2	3,130	10.43%	2,177	32.81%	69.55%
B: PAY_0 > 0	6,818	22.73%	3,429	51.67%	50.29%

Increment: 3,688 additional clients (+117.83% versus A), 1,252 additional positive labels and +18.87 percentage points of coverage. The added population has a 33.95% label rate. Coverage is not loss prevention.

RECOMMENDATIONS AND BOUNDARIES

Clarify status codes and the outcome definition; test both scopes on recent independent data; select a scope only after assessing capacity, costs and the follow-up action. Do not deprioritise PAY_0 >= 2 clients solely because their limits are higher.

This study describes a 2005 Taiwan sample. Causality, future performance and intervention benefits are not established. No immediate lending-policy change or current Canadian risk estimate is supported.

Credit Limits: Testing the Overall Pattern

Credit-limit bands use sample quartiles of TWD 50,000, 140,000 and 240,000. Each upper endpoint is included. These exploratory bands are not bank policy thresholds.

OVERALL RESULTS BY CREDIT-LIMIT BAND

Credit limit (TWD)	Clients	Positive labels	Label rate	Coverage
<= 50,000	7,676	2,440	31.79%	36.77%
50,000-140,000	7,614	1,882	24.72%	28.36%
140,000-240,000	7,643	1,326	17.35%	19.98%
> 240,000	7,067	988	13.98%	14.89%

The lowest band has a 31.79% label rate versus 13.98% in the highest. However, PAY_0 >= 2 represents 17.18% of the lowest band and only 4.19% of the highest, suggesting that customer mix contributes to the crude pattern.

WITHIN-STATUS COMPARISON

Credit limit (TWD)	PAY_0 <= 0	PAY_0 = 1	PAY_0 >= 2
<= 50,000	19.93% n = 5,183	42.25% n = 1,174	69.07% n = 1,319
50,000-140,000	14.73% n = 5,681	35.35% n = 911	70.74% n = 1,022
140,000-240,000	11.79% n = 6,243	27.78% n = 907	68.56% n = 493
> 240,000	9.89% n = 6,075	26.15% n = 696	69.26% n = 296

Cells show within-group label rates; n denotes client count.

INTERPRETATION

Rates still decline with increasing limits within PAY_0 <= 0 and = 1. Composition across the three broad status groups therefore does not fully explain the overall relationship.

Within PAY_0 >= 2, rates range from 68.56% to 70.74% without a clear monotonic pattern. This does not establish statistical equivalence or imply that raising limits reduces risk. Other customer differences remain uncontrolled.

Analysis Basis, Validation & Limitations

DATA AND METRICS

Analysis basis: the submitted notebook8b9c4b2425.ipynb, using Kaggle UCI_Credit_Card.csv. Six months of Apr-Sep 2005 histories are present, but the principal segmentation uses September PAY_0. No cross-month transition or delay-frequency conclusion is made.

Label rate: group positive labels / group clients. **Workload share:** group clients / 30,000. **Coverage:** group positives / 6,636. Scope B includes A; the increment is PAY_0 = 1. Label 0 does not mean a client never had a payment delay.

DATA QUALITY AND DELIVERY CHECKS

Check	Result and treatment
Population	30,000 unique IDs; 0 missing cells; 0 fully duplicated rows.
Category codes	345 education and 54 marriage values outside the reviewed dictionary; retained.
PAY codes	-2 and 0 remain unresolved. Code 1 differs sharply across months; temporal comparability is unverified.
Numerical verification	Report metrics independently recalculated from the original UCI workbook and matched to saved notebook results.
Notebook execution	Section 6 references undefined limit_analysis; change to limit_summary, then restart and run all. Saved PASS output is not proof of a clean run.

MATERIAL LIMITATIONS

The historical selected sample does not represent current Canadian risk. The target threshold is incompletely documented and is not equated with regulatory or 90+ DPD default. Candidate scopes were assessed in-sample, without independent or out-of-time validation. Stratified results remain descriptive. Review costs, intervention outcomes and losses are unavailable; a credit limit is not an outstanding balance or realised loss.

SOURCES

[1] Yeh, I. (2009). *Default of Credit Card Clients*. UCI Machine Learning Repository. doi.org/10.24432/C55S3H. CC BY 4.0.
[2] Kaggle: [Default of Credit Card Clients](#).
[3] Kailai Yang, notebook8b9c4b2425.ipynb, submitted analysis reviewed 09 September 2026.